

Passivity-Based Skill Motion Learning in Stiffness-Adaptive Unified Force-Impedance Control

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Abstract—Tactile robots shall be deployed for dynamic task execution in production lines with small batch sizes. Therefore, these robots should have the ability to respond to changing conditions and be easy to (re-)program. Operating under uncertain environments requires unifying subsystems such as robot motion and force policy into one framework, referred to as tactile skills. In this paper, we propose the enhancement of these skills for passivity-based skill motion learning in *stiffness-adaptive unified force-impedance control*. To achieve the increased level of adaptability, we represent all tactile skills by three basic primitives: contact initiation, manipulation, and contact termination. To ensure passivity and stability, we develop an energy-based approach for unified force-impedance control that allows humans to teach the robot motion through physical interaction during the execution of a tactile task. We incorporate our proposed framework into a tactile robot to experimentally validate the motion adaptation by interaction performance and stability of the control. While the polishing task is presented as our use case through the paper, the experiments can also be carried out with various tactile skills. Finally, the results show the novel controller's stability and passivity to contact-loss and stiffness adaptation, leading to successful programming by interaction.

I. INTRODUCTION

With the demand for customized production, flexible production lines for small batch sizes are required [1]. These production lines ask for flexible adaptation to varying conditions [2]. Traditional position-controlled robots do not provide these capabilities. Compliant interaction for robots only become feasible, e.g., when using admittance control [3], impedance control [4], or force control [5]. Depending on the type of the robot and the controller, the tactile performance may vary significantly [6].

Highly tactile robots enable human-like intelligent recognition of touch and can perform sensitive tasks such as polishing, screwing, levering, or cutting [7]. Ideally, with those robots, the programmer only needs to command the robot what task to perform and where to perform it to (re-)program the robot task [8]. For example, when polishing cars, the human would tell the robot to polish in a certain spot and then quickly show it the next soiled spot while the robot is still in motion. Just as one human would show another where the polishing task is most required.

However, with the current state of the art in tactile robot programming, fairly accurate positioning is required before

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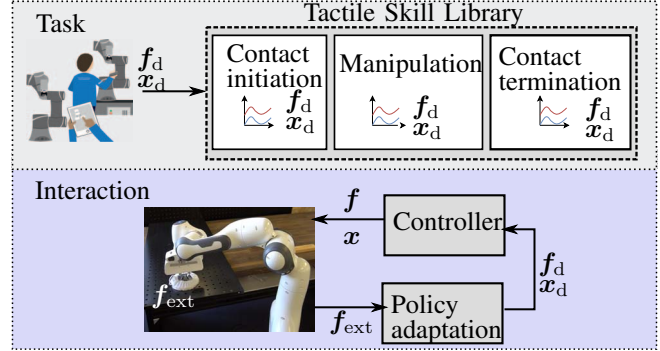


Fig. 1: **Adaptive tactile skills.** The tactile skill library provides the basic primitives to perform tactile robot tasks. In reality, however, actual situations differ from those defined in the library. Therefore, methods are required to extend the library and allow robots to adapt these well-defined skills to a new environment.

the tactile task can be successfully performed [9]. This requires expert knowledge and inefficient robot downtime to (re-)program the robot action [10]. Therefore, to enable flexible adaptation of tactile robot capabilities, new concepts are required to simplify the (re-)programming of robot tasks.

In this paper, a *tactile skill motion adaptation through unified force-impedance control* is proposed, as schematically depicted by Fig. 1. Three tactile skill primitives are defined, which are valid for any tactile robot-environment interaction; namely contact initiation, manipulation, and contact termination. To enable robot (re-)programming by interaction, we consider the manipulation primitive and design an adaptive unified force-impedance controller. This controller couples a stiffness-adaptive impedance control with force control to passively adapt the manipulation primitive's periodic motion and direction of action. The result is a novel method applicable to any tactile robot skill, as it is illustrated in the accompanying video. For the sake of clarity in the paper we focus our explanation on the polishing task which is demonstrated using a Franka Emika (FE) robot arm.

The paper is structured as follows. In Sec. II, we summarize the state of the art. Section III introduces our proposed framework for stiffness-adaptive tactile skills. The experimental procedure as well as the results are presented in Sec. IV and Sec. V. Finally, Sec. VI concludes the paper.

II. STATE OF THE ART

Robotic tactile skills such as polishing, screwing, unscrewing, and levering require precise control of interaction at the end-effector. In other words, tactile skills should contain a motion generation unit with force policy. There are different perspectives in the literature for combining

motion generation and force control in robotic manipulators. For instance, Zielenski et al. [11] studied a method that classifies the manipulator behavior under three different phases: free-motion where the force is of no significance, exerting generalized forces, and lastly the transitions between the latter two behaviors. Robot manipulators need to develop complex perceptuomotor skills, as they further challenge real-world problems [12]. Therefore, defining tactile skills with a position control strategy alone is an impractical solution, and other control methods should be considered instead [4]. Furthermore, force control has been considered in many works, where the proposed controllers are validated based on constant force values or as thresholds/constraints [13]–[17]. However, the robots need to be robust enough to autonomously operate in unstructured environments and perform desired tasks in the presence of perception uncertainties [18].

Impedance control is a well-known approach that imposes a dynamic behavior between the external interaction and the desired motion, rather than tracking motion or force trajectory independently [4]. These dynamics can be realized in the joint space, operational space, or even in the redundant space of a robot manipulator [19]. The impedance inertia, damping, and stiffness parameters are usually selected constant in each direction based on the assigned tasks. The impedance dynamics represent a passive mapping between the external force input and the motion of the robot as an output. This ensures that the system does not generate additional energy when interacting with the passive environment and can therefore be considered stable [20]. However, adaptive change of the impedance parameters seems beneficial as well in many applications to impose a more wise and human-like behavior for the external interaction on the EEF [21]. Nevertheless, the variation of impedance parameters compromises the passivity of the control loop and therefore leads to instability. To this end, the concept of the energy tank is exploited to alter the dynamics of the closed-loop system and ensure passivity [22]. To include compliant force tracking on the environment with unknown geometry, the Wavelet Neural Network (WNN) has effectively been used in [23]. The variable admittance approaches can also be used to coordinate the human and robot motion in applications with shared autonomy [24]. However, still, with the state of the art in tactile robot programming, rather accurate positioning needs to be ensured before the task is performed.

In this work, we demonstrate that the required skill motion learning can be performed by adapting stiffness within unified force-impedance control, leveraging the energy transfer between robot and environment. We extend the unified force-impedance control paradigm by featuring control shaping function dynamics to allow humans to adapt any tactile skill by physical interaction.

III. METHODOLOGY

Our framework is based on adaptive force-impedance control that can be used for dynamic programming of tactile skill motions by interaction. To this end, we use stiffness

adaptation to enable programming by interaction and define tactile skill primitives for motion learning.

The taxonomy for tactile skills is first applied at the primitive level, and all the tactile skills are defined by three basic primitives. Unified force-impedance control is then extended to introduce the stiffness-adaptive dynamics to the control shaping function. This allows humans to adapt each tactile skill based on the energy transfer within the system. The proposed framework is illustrated in Fig 2 and explained in detail in the following.

A. Tactile primitives

To ensure compatibility with high-level planning and low-level control algorithms, any tactile skill is decomposed into three primitives. Each primitive $\mathcal{P}_i$, contributed to the desired skill, has its own policy $(\mathbf{x}_d, \mathbf{f}_d)$ and, therefore, force-motion generation unit:

- $\mathcal{P}_1$ – Contact initiation between tool and the work piece,
- $\mathcal{P}_2$ – Manipulation defined by a periodic motion and corresponding force policy,
- $\mathcal{P}_3$ – Contact termination to stop the interaction and decide if the desired skill is achieved.

The transition parameters between these primitives can be derived automatically from the perception of the scene through exteroceptive sensing and external user interaction, or manually through predefined scenarios. A blending strategy is then exploited for seamless integration of the primitives to achieve the desired skills.

In this framework, $\mathcal{P}_2$ is the core of any skill. It is defined by a periodic Dynamic Movement Primitive (DMP) as motion generator. The DMP is composed of the frequency and shape of the motion, and the required force policy is designed to be constant in the direction of motion. For adaptation of the motion, appropriate shaping functions are modelled to allow human interaction and safe loss-of-contact behavior, respectively, as described detailed in the following.

B. Stiffness-adaptive unified force-impedance control

The proposed control law for adaptive tactile skills is extended from the unified force-impedance controller introduced in [25] and comprises four main components:

- I) tracking the desired motion $\mathbf{x}_d$,
- II) initializing and controlling the desired force $\mathbf{f}_d$,
- III) deciding on the robot stiffness level ρ_{imp} , and
- IV) gravity compensation τ_g .

The robot dynamics equation in Cartesian space can be expressed as

$$\mathbf{M}_C(\mathbf{q})\ddot{\mathbf{x}} + \mathbf{C}_C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{x}} + \mathbf{g}_C(\mathbf{q}) = \mathbf{f}_{\text{in}} + \mathbf{f}_{\text{ext}}, \quad (1)$$

$$\boldsymbol{\tau}_{\text{in}} = \mathbf{J}^T(\mathbf{q})\mathbf{f}_{\text{in}}, \quad (2)$$

$$\boldsymbol{\tau}_g = \mathbf{J}^T(\mathbf{q})\mathbf{g}_C(\mathbf{q}), \quad (3)$$

where the robot mass matrix, Coriolis/centrifugal matrix, Jacobian matrix, and gravity vector in Cartesian space are denoted as $\mathbf{M}_C(\mathbf{q}), \mathbf{C}_C(\mathbf{q}, \dot{\mathbf{q}}), \mathbf{J}(\mathbf{q}) \in \mathbb{R}^{6 \times 6}$ and $\mathbf{g}_C(\mathbf{q})$ respectively. Furthermore, $\boldsymbol{\tau}_{\text{in}} \in \mathbb{R}^n$ is the control input

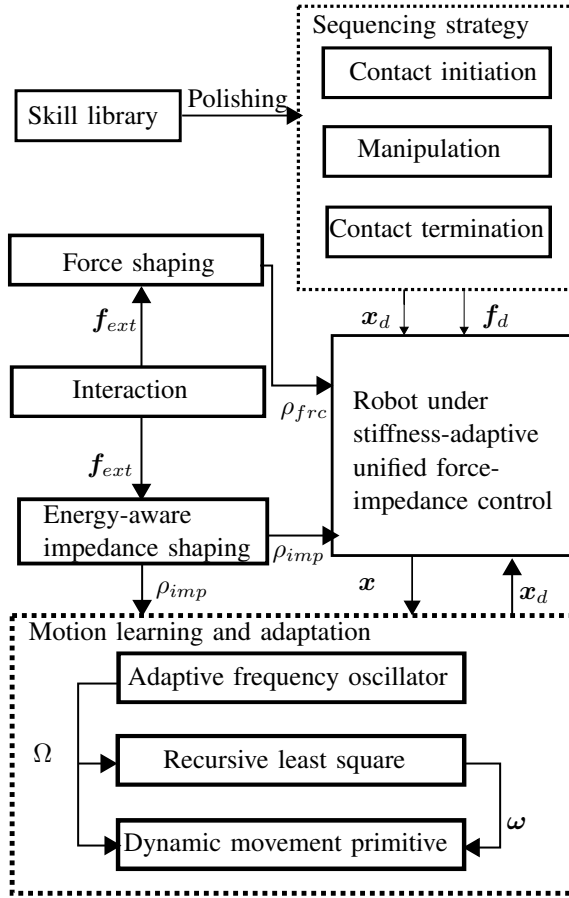


Fig. 2: **Block diagram for stiffness-adaptive tactile skill architecture.** Tactile skills are achieved by combining three basic primitives. The core of any skill is the manipulation primitive defined by a periodic motion and a desired force in the direction of action. The controller shaping functions passively regulate the interaction with the environment, such that ρ_{imp} allows motion adaptation by interaction, and ρ_{frc} is responsible for maintaining the contact force.

torque and $\mathbf{f}_{ext} \in \mathbb{R}^6$ is the external wrench acting on the robot.

The input torque, τ_{in} , is:

$$\tau_{in} = \tau_{imp} + \tau_{frc} + \tau_g, \quad (4)$$

where τ_{imp} , τ_{frc} , and $\tau_g \in \mathbb{R}^n$ are the input torques for the impedance, force, and gravity respectively.

1) *Force controller:* To initiate and control a desired contact force $\mathbf{f}_d$ at the end-effector according to the external force, $\mathbf{f}_{ext}$, the force control is defined as

$$\tau_{frc} = \rho_{frc} \mathbf{J}^T(\mathbf{q}) \mathbf{f}_{frc}, \quad (5)$$

$$\mathbf{f}_{frc} = \mathbf{f}_d + k_p \tilde{\mathbf{f}}_{ext} + k_i \int \tilde{\mathbf{f}}_{ext} dt + k_d \dot{\tilde{\mathbf{f}}}_{ext}, \quad (6)$$

$$\tilde{\mathbf{f}}_{ext} = \mathbf{f}_d - \mathbf{f}_{ext}, \quad (7)$$

where $\mathbf{f}_{frc} \in \mathbb{R}^6$ is the feedback and feed forward force term, and k_p, k_i, k_d are PID gains. The force shaping function ρ_{frc} in (5) deactivates the force controller, when the robot deviates from the set-point of the end-effector. The idea behind the shaping function is to avoid undesired motions, especially when contact loss occurs. Therefore, ρ_{frc} is designed with

three cases based on the deviation from the set-point

$$\tilde{\mathbf{x}} = \mathbf{x}_d - \mathbf{x}. \quad (8)$$

The first case resembles the condition before contact where $\tilde{x}_z > 0$ and, thus, the energetic potential $\mathbf{f}_d^T \tilde{\mathbf{x}} \geq 0$. The second condition is $\tilde{x}_z < 0$ where the desired position is violated and thus the energetic potential turns $\mathbf{f}_d^T \tilde{\mathbf{x}} < 0$. To ensure a transition between both situations and prevent errors due to a mismatch of force and motion direction, a third case is included, which allows smooth reduction of the control torque within a certain position range up to a threshold $\delta_{frc} > 0$. This happens for example when the robot loses the contact with a surface during polishing and falls in z-direction. Finally, ρ_{frc} becomes

$$\rho_{frc} = \begin{cases} 1, & \tilde{x}_z \geq 0 \\ 0.5(1 + \cos(\pi(\frac{|\tilde{x}_z|}{\delta_{frc}}))), & 0 < |\tilde{x}_z| \leq \delta_{frc} \\ 0 & else \end{cases} \quad (9)$$

2) *Energy-aware impedance controller:* To introduce an impedance behavior to the end-effector and allow human-based skill adaptation, the following control law is applied,

$$\tau_{imp} = \mathbf{J}^T(\mathbf{q})(\mathbf{K}_C \tilde{\mathbf{x}} + \mathbf{D}_C \dot{\tilde{\mathbf{x}}} + \mathbf{M}_C(\mathbf{q}) \ddot{\mathbf{x}}_d + \mathbf{C}_C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{x}}_d). \quad (10)$$

The stiffness and damping matrices are denoted by $\mathbf{K}_C$ and $\mathbf{D}_C \in \mathbb{R}^{6 \times 6}$, respectively. The energy storage in the closed loop system is operated by

$$S = \frac{1}{2} \dot{\tilde{\mathbf{x}}}^T \mathbf{M}_C(\mathbf{q}) \dot{\tilde{\mathbf{x}}} + \frac{1}{2} \tilde{\mathbf{x}}^T \mathbf{K}_C \tilde{\mathbf{x}}. \quad (11)$$

Considering (1), (5), and (10), the time derivative of (11) is

$$\dot{S} = \dot{\tilde{\mathbf{x}}}^T (-\mathbf{C}_C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\tilde{\mathbf{x}}} - \mathbf{D}_C \dot{\tilde{\mathbf{x}}} - \mathbf{K}_C \tilde{\mathbf{x}} - \mathbf{f}_{frc} - \mathbf{f}_{ext}) + \frac{1}{2} \dot{\tilde{\mathbf{x}}}^T \dot{\mathbf{M}}_C(\mathbf{q}) \dot{\tilde{\mathbf{x}}} + \dot{\tilde{\mathbf{x}}}^T \mathbf{K}_C \tilde{\mathbf{x}} + \frac{1}{2} \tilde{\mathbf{x}}^T \dot{\mathbf{K}}_C \tilde{\mathbf{x}}, \quad (12)$$

where the closed loop equation,

$$\mathbf{M}_C(\mathbf{q}) \ddot{\tilde{\mathbf{x}}} = -\mathbf{C}_C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\tilde{\mathbf{x}}} - \mathbf{D}_C \dot{\tilde{\mathbf{x}}} - \mathbf{K}_C \tilde{\mathbf{x}} - \mathbf{f}_{frc} - \mathbf{f}_{ext}. \quad (13)$$

has been used. Note that the skew-symmetry of the matrix $(\dot{\mathbf{M}}_C(\mathbf{q}) - 2\mathbf{C}_C(\mathbf{q}, \dot{\mathbf{q}}))$ is also applied. Considering (12) and the positive-definiteness of the damping matrix $\mathbf{D}_C$, it can be deduced that

$$\dot{S} \leq \frac{1}{2} \tilde{\mathbf{x}}^T \dot{\mathbf{K}}_C \tilde{\mathbf{x}} - \dot{\tilde{\mathbf{x}}}^T \mathbf{f}_{frc} - \dot{\tilde{\mathbf{x}}}^T \mathbf{f}_{ext} + \tilde{\mathbf{x}}^T \dot{\mathbf{f}}_{ext}. \quad (14)$$

$$P_{acv, imp} = -\dot{\tilde{\mathbf{x}}}^T \mathbf{f}_{ext}, \quad (15)$$

$$P_{acv, K} = \frac{1}{2} \tilde{\mathbf{x}}^T \dot{\mathbf{K}}_C \tilde{\mathbf{x}}, \quad (16)$$

$$P_{acv, frc} = -\dot{\tilde{\mathbf{x}}}^T \mathbf{f}_{frc}, \quad (17)$$

$$P_{ext} = \tilde{\mathbf{x}}^T \dot{\mathbf{f}}_{ext}, \quad (18)$$

shows the passivity with respect to the summation of the non-passive (i.e., active) control power P_{acv} as well as the external power P_{ext} .

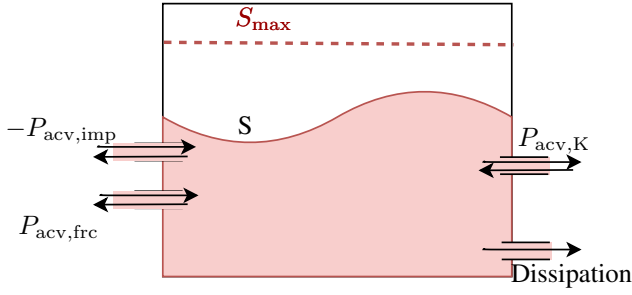


Fig. 3: **Energy transfer scheme between robot and the environment.** The energy is bounded by $S_{\max}$, such that the stability of the system during adaptation by interaction is preserved.

Here, we propose applying the derived energy models to bound the energy transferred from the robot to the human and to modulate the reaction of the robot to the human interruption (see Fig. 3). In other words, the stiffness of the robot changes with the shaping function, $\rho_{\text{imp}} \in [0, 1]$, where the stiffness adaptation rate, $\dot{\rho}_{\text{imp}}$, is formed based on the energy storage function of the robot, S , and the power inputs to the system $\langle -\dot{x}_d, f_{\text{ext}} \rangle$, $\langle \dot{x}, K_C \rangle$ by the force-motion policy generator. Inspired by [26], the stiffness adaptation is defined as

$$K_C = \rho_{\text{imp}}(t) K_{\max}, \quad (19)$$

and the adaptation parameter ρ_{imp} is obtained by

$$\dot{\rho}_{\text{imp}} = \begin{cases} \min\{\rho, 0\}, & \rho_{\text{imp}} = 1 \\ \rho, & 0 < \rho_{\text{imp}} < 1, \rho_{\text{imp}}(0) = 0, \\ \max\{\rho, 0\}, & \rho_{\text{imp}} = 0 \end{cases} \quad (20)$$

where, ρ is given by

$$\rho = h\rho_{\text{imp}} + \rho_{\min}, \quad (21)$$

and,

$$h = \frac{S_{\max} - S}{S_{\max}}. \quad (22)$$

Note that, in order to have an initial increment for the case $\rho_{\text{imp}} = 0$, a small positive constant $\rho_{\min}$ has been introduced into the shaping function dynamics.

The above dynamic ensures that ρ_{imp} acts as a decreasing or an increasing exponential function, for $S > S_{\max}$ and $S < S_{\max}$, respectively. Overall, this leads to a stiffness behavior as follows.

$$K_C = \begin{cases} \text{fully autonomous,} & \rho_{\text{imp}} = 1 \wedge S < S_{\max} \\ \text{fully loose,} & S > S_{\max} \\ \text{shared autonomy,} & \rho_{\text{imp}} = 0 \wedge S < S_{\max} \end{cases} \quad (23)$$

After S reaches the maximum allowable energy threshold $S_{\max}$, energy shaping starts¹.

When the robot operates without any human interruption, i.e., $\rho_{\text{imp}} = 1$, the stiffness equals its maximum value $K_{\max}$. During human intervention, i.e., $P_{\text{acv,imp}} < 0$, the threshold

¹For now, $S_{\max}$ is set heuristically by the user. Its autonomous adaptation is left for future studies.

is exceeded since the human inserts energy to the system. As the robot adapts its motion trajectory based on the human guidance, $\dot{\rho}_{\text{imp}}$ increases in (20), resulting in progressive energy dissipation.

C. Force-motion policy

As mentioned before, the primitive $\mathcal{P}_2$ is defined as the core of any tactile skill which is constituted by a periodic motion and a matching force policy. A periodic motion is comprised by its frequency Ω and shape ω . Moreover, the motion generator is coupled to the controller via a shaping function ρ_{imp} to allow autonomous adaptation and teaching of new skill motions by the human. However, for sake of clarity, we assume the desired force to be constant during contact, so we concentrate on the developed energy-awareness within our system on motion and stiffness adaptation level.

1) *Adaptive frequency oscillators*: In order to extract the frequency from a motion shown by the human operator, we employ the online learning and modulation method for periodic movements introduced in [27], which is shortly reviewed in the following. Generally, the core of the algorithm is formed by the dynamics [26]

$$\dot{\theta} = (1 - \rho_{\text{imp}})(\phi - aE \sin(\theta)), \quad (24)$$

$$\dot{\phi} = -(1 - \rho_{\text{imp}})(aE \sin(\theta)), \quad (25)$$

$$E = \text{diag}(x - \hat{x}), \quad (26)$$

where $a \in \mathbb{R}$ is the coupling constant and error matrix $E \in \mathbb{R}^{m \times m}$ refers between the current position x and predicted motion $\hat{x}$, $(1 - \rho_{\text{imp}})$ introduces the progressive learning and adaptation behavior to the system, such that when the robot is fully autonomous $\rho_{\text{imp}} = 1$, the adaptation stops. The frequency vector ϕ and the corresponding phase vector θ belong to $\mathbb{R}^m$, where m is the number of task space coordinates to be adapted. The predicted motion is computed by

$$\hat{x}_i = \sum_{c=0}^M (\alpha_{i,c} \cos(c\theta_i) + \beta_{i,c} \sin(c\theta_i)), \quad i = 1, 2, \dots, m, \quad (27)$$

where M denotes the number of Fourier components. The amplitudes $\alpha_{i,c}$, $\beta_{i,c}$ are updated as follows.

$$\begin{bmatrix} \dot{\alpha}_{i,c} \\ \dot{\beta}_{i,c} \end{bmatrix} = \begin{bmatrix} (1 - \rho_{\text{imp}}) \mu \cos(c\theta_i) e_i \\ (1 - \rho_{\text{imp}}) \mu \sin(c\theta_i) e_i \end{bmatrix}. \quad (28)$$

The error e_i is the i^{th} diagonal element of E , and μ is the learning constant. Finally, the frequency Ω of the motion generator is set to the minimum element of ϕ [28].

In this work, M , μ , and α are chosen to be 1, 1, and 50, respectively. The initial value for the phase vector is $\theta_0 = 2\pi \mathbf{I}_{m \times 1}$.

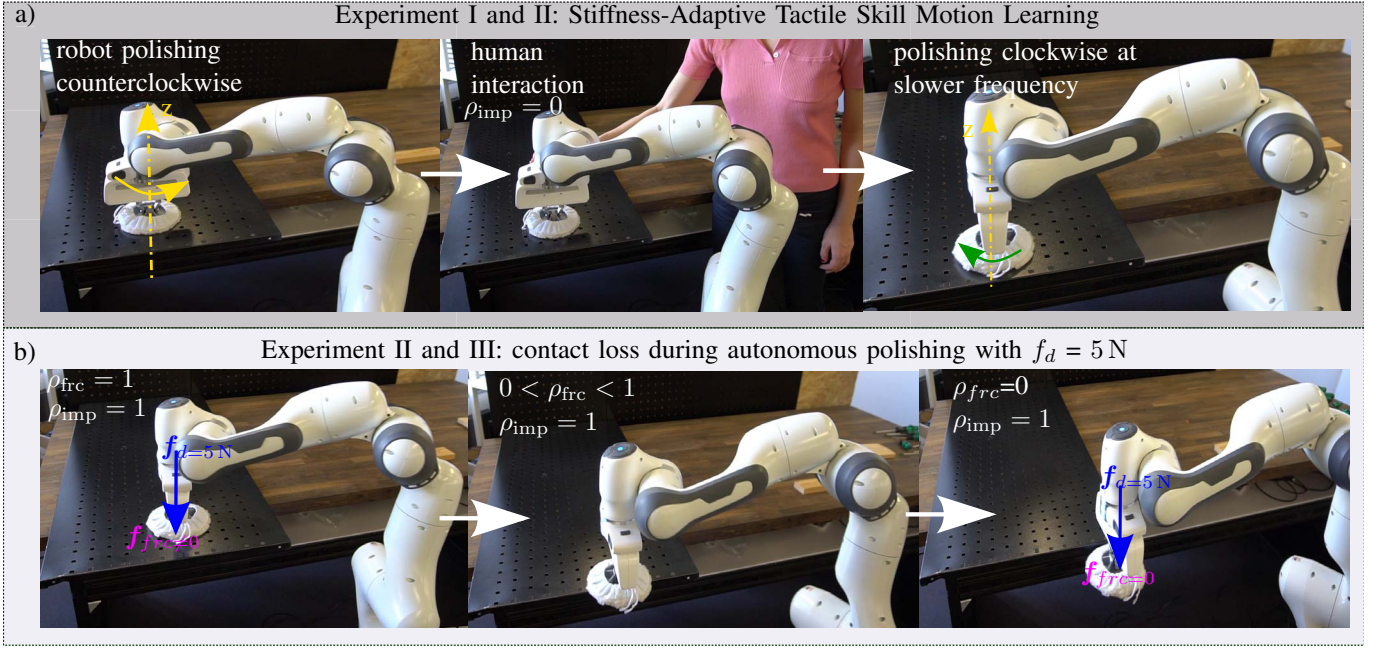


Fig. 4: **Tactile robot polishing a surface.** Image series for the experimental procedure a) Experiment I and II for passivity-based stiffness-adaptation by human interaction: human adapts the speed and direction of the free and contact motion b) Experiment III shows the performance of force and impedance shaping functions together: the robot autonomously and smoothly stops the force controller after contact loss.

2) *Motion generator:* DMPs have been designed to represent a smooth robot motion [29]. A periodic DMP controlling the motion of one degree of freedom $m = 1$, x_d is given by a second order differential equation system

$$\dot{s} = \Omega, \quad (29)$$

$$\dot{z} = \Omega(\alpha_z(\beta_z(x_g - x_d) - z) + \gamma(s)), \quad (30)$$

$$\dot{x} = \Omega z, \quad (31)$$

where x_g the attractor point and $s \in R$ is the phase variable which is used to make the system time independent. Furthermore, the constant parameters $\alpha_z, \beta_z > 0$ are selected such that the system has a unique attractor x_g . The nonlinear forcing term $\gamma(s)$ is defined as

$$\gamma(s) = \frac{\sum_{i=1}^N \omega_i \psi_i(s)}{\sum_{i=1}^N \psi_i(s)}, \quad (32)$$

$$\psi_i(s) = \exp(h(\cos(s - c_i))). \quad (33)$$

This allows us to freely design periodic trajectories from an initial position x_0 to the goal x_g by linear combination of N radial-basis functions $\psi_i(s)$. c_i is the center of the functions and h determines their width. In this work, we choose $N = 30$, $\alpha_z = 20$, $\beta_z = 5$, $h = 2.5$, and $c_i = \frac{2\pi}{N}(i - 1)$. The frequency Ω is chosen based on adaptive frequency oscillators. The generated trajectory with $x_d, \dot{x}_d$, and $\ddot{x}_d$ is finally obtained by Euler integration of (31) with the initial values $s = 0$, $x = x_0$, and $\dot{x} = \Omega z = 0$. To extract the shape of the motion namely the weight matrix $\omega \in \mathbb{R}^{m \times N}$, each weight vector $\omega_i \in \mathbb{R}^m$ where $m = 3$ is

updated by the recursive least squares as follows [27]

$$\omega_i(t+1) = \omega_i(t) + \psi_i \text{diag}(\mathbf{P}_i(t+1)) e_{r_i}(t), \quad (34)$$

$$\mathbf{P}_{j,i}(t+1) = \frac{1}{\lambda} \left(\mathbf{P}_{j,i}(t) - \frac{\mathbf{P}_{j,i}(t)^2}{\frac{\lambda}{\psi_i} + \mathbf{P}_{j,i}(t)} \right), \quad (35)$$

$$e_{r_i}(t) = (1 - \rho_{imp})(\gamma_d(t) - \omega_i), \quad (36)$$

$$\mathbf{P}_{j,i}(0) = 1, \quad (37)$$

where γ_d the desired trajectory shape demonstrated by the human [29]. The inverse covariance vector $\mathbf{P}_i \in \mathbb{R}^m$ ($i = 1, 2, \dots, N; j = 1, 2, \dots, m$) corresponding to the weights ω_i is computed by using the forgetting factor $\lambda = 0.9995$.

When the impedance shaping function ρ_{imp} , reaches 1, the adaptation of the shape of the motion in (37) stops and is accepted as successfully approximated to the desired shape.

3) *Force policy:* the force policy is especially useful to mitigate the chattering/jamming behavior of the robot while trying to maintain the contact force and the pose error is high due to friction, etc. In this work, the force policy is chosen to be constant at the direction of action and coupled to the controller shaping function, ρ_{fr} .

IV. EXPERIMENTAL PROCEDURE

To demonstrate the real-life performance of our framework, the experiments are conducted using a Franka Emika robot for a polishing use case (see Fig. 4). The experiments were conducted considering three different cases:

- I) passivity-based learning by interaction during free motion in adaptive unified force-impedance control,
- II) adaptation by interaction during contact to validate the overall adaptive tactile skill definition,

III) performance of force and impedance shaping simultaneously.

To achieve robust and flexible behavior and to avoid long run times, the primitives are connected using a sequencing strategy [30], resulting in seamless and smooth integration of the primitives into the desired skill. Additionally, the skill definition enables straightforward and partly autonomous programming of complex skills. Algorithm 1 illustrates the procedure for an example scenario of polishing skill.

Algorithm 1 Skill definition for polishing

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1: procedure POLISHING
2:   parameter server  $\leftarrow x_0, x_g, f_d, \omega, \Omega, \rho_{\text{frc}}, \rho_{\text{imp}}$ 
3:   contact initiation  $\mathcal{P}_1 \leftarrow x_g$ 
4:   manipulation  $\mathcal{P}_2 \leftarrow x_0, x_g, \Omega, \omega, f_d, \rho_{\text{frc}}, \rho_{\text{imp}}$ 
5:   contact termination  $\mathcal{P}_3 \leftarrow x_g$ 
6:  $\mathcal{P}_1$ :
7:   if  $x_g = x$  then
8:     go to  $\mathcal{P}_2$   $\triangleright x_0 = x$ 
9:   end if
10:  $\mathcal{P}_2$ :
11:   if  $\rho_{\text{frc}} < 1$  then
12:     go to  $\mathcal{P}_1$   $\triangleright x_g = x \mid f_{\text{ext}} \neq 0$ 
13:   end if
14:   if  $\rho_{\text{imp}} < 1$  then
15:     adapt  $\Omega, \omega$ 
16:   end if
17:   if  $x = x_g$  then
18:     go to  $\mathcal{P}_3$   $\triangleright x_0 = x, \rho_{\text{frc}} = 1$ 
19:   end if
20:  $\mathcal{P}_3$ :
21:   if  $x = x_g$  then
22:     return success
23:   end if
24: end procedure

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Any tactile skill such as polishing is defined by combining the above primitives (contact initiation, manipulation, and contact termination). Each of the primitives $\mathcal{P}_i(x_d, f_d)$ comprises a DMP and force policy. Furthermore, the DMP equation in (31) has the initial position x_0 and the goal x_g , which are varied between primitive skills and depends on the sequencing strategy. To solve the dependency problems, initial position x_0 and goal x_g are updated and combined according to Algorithm 1. The idea behind this operation is that when one primitive executes, this primitive policy is updated to generate the motion x_d and force trajectory f_d .

A. Experiment I

In the first experiment, the robot starts to perform a circular motion autonomously $\rho_{\text{imp}} = 1$ and a human operator interacts with the robot to vary the motion speed. Impedance shaping ρ_{imp} activates the adaptation of the motion based on the energy transfer between the robot and its environment, as bounded by the maximum allowable energy $S_{\text{max}} = 14$ J. The adaptation of the motion frequency Ω and x_d vs x

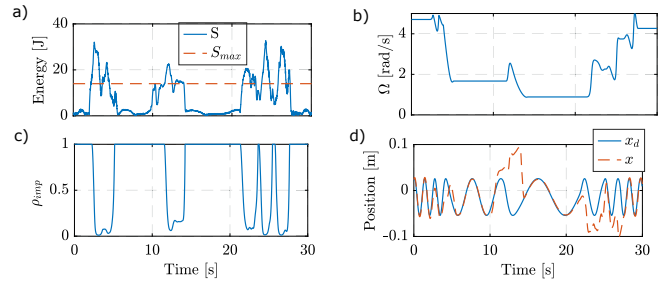


Fig. 5: Experiment I: Evolution of S , Ω and ρ_{imp} in free motion during adaptation of x_d a) and c) ρ_{imp} activates the adaptation of the motion based on the energy transfer between robot and the environment bounded by $S_{\text{max}} = 14$ J. b) The adaptation of Ω d) The adaptation of motion and comparison between the current state x_d and x based on the energy transfer.

are evaluated based on the energy transfer and impedance shaping.

B. Experiment II

In the second set of experiments, the robot polishes a surface and autonomously performs the entire stiffness-adaptive tactile skill definition, starting with the contact initiation primitive. Then, the manipulation primitive is activated to perform the polishing process by simultaneous force application and motion. The robot applies the desired force $f_d = 5$ N to the surface and executes the desired trajectory. During the primitive manipulation of the polishing process, the human operator interacts with the robot to adjust the speed and shape of the executed motion. To allow the human operator to interact with the robot during the contact skill, the maximum energy bound is set to $S_{\text{max}} = 2$ J.

C. Experiment III

In the last set of experiments, the performance of the force shaping function ρ_{frc} during the contact-loss is validated. The robot autonomously executes the entire framework, robustly stopping and restoring contact via ρ_{frc} . The applied force in the direction of action $f_{\text{frc},z}$ is expected to change uniformly from 5 N to 0 N after contact loss.

V. RESULTS AND DISCUSSION

The results for Experiment I in Fig. 5.a) and Fig. 5.c) show that the impedance shaping function ρ_{imp} is activated whenever the energy transfer between humans and robots is above the predefined maximum energy level ($S_{\text{max}} = 14$ J). More specifically, right after the interaction ρ_{imp} decreases to 0 from 1 such that the stiffness is also zero ($K_C = 0$), allowing humans to easily interact. Finally, the frequency of the motion is adapted during the interaction with respect to the guided motion. Additionally, the comparison between the desired trajectory x_d computed by the motion generator and the current state of the robot x is depicted in Fig. 5.b) and Fig. 5.d). Based on the results of the motion comparison, the human operator is able to slow down the robot during intervention and then speed it up again.

In Experiment II, it can be seen that in Fig. 6.a) and Fig. 6.b), where the S is greater than S_{max} , the human operator could change the shape and the frequency of the

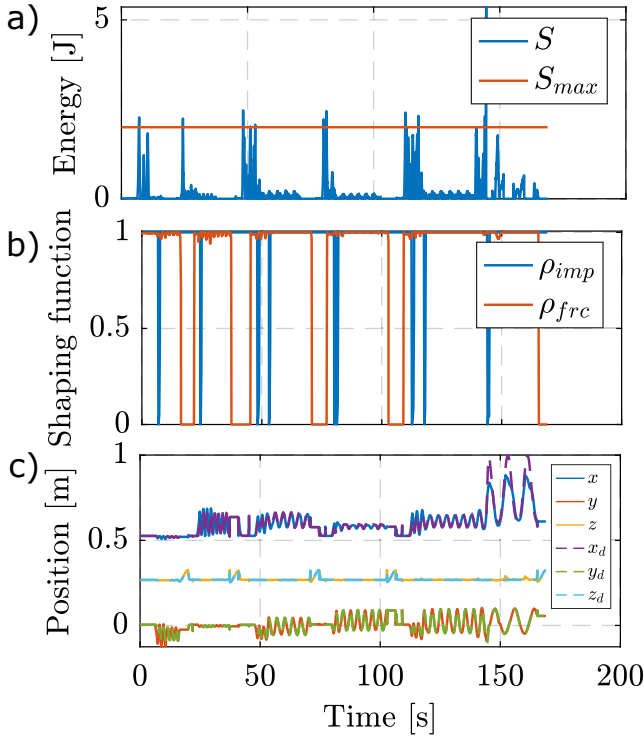


Fig. 6: **Experiment II: Evolution of S and ρ_{imp} , adaptation of x_d and y_d in contact with the environment.** S_{max} is set to 2 J to enable the human to interact with the robot that polishes by applying $f_d = 5$ N on the surface.

motion. As expected, after interacting with the human, the robot autonomously performs the rest of the adapted skill using the contact initiation and termination primitives.

Finally, in Experiment III, the robot robustly and safely stops the force controller by smoothly changing ρ_{frc} from one to zero as shown in Fig. 7.a). Consequently, the applied force to the surface, f_z , decrease to zero from 5 N as depicted in Fig. 7.b) and Fig. 7.d). Note that the robot is fully autonomous during the force regulation, as can be seen from the fact that $\rho_{imp} = 1$. Additionally, once contact with the surface is restored, ρ_{frc} is equal to 1 again and the manipulation primitive continues to apply force and executes the motion trajectory in parallel until another contact loss occurs.

VI. CONCLUSION

In this study, we developed a framework for skill motion learning in stiffness-adaptive unified force-impedance control based on energy transfer between robot and environment. The tactile capabilities are partitioned into three basic primitives, contact initiation, manipulation, and contact termination. The passivity-based dynamics of impedance shaping function is then introduced into unified force-impedance control, which results in motion adaptation capabilities by human interaction. We demonstrated the performance and stability of our system in real-world experiments, such as polishing a surface, unscrewing a knob, and levering a lid. In this paper, the polishing task is presented, while in the attached video the transition to other tasks is demonstrated

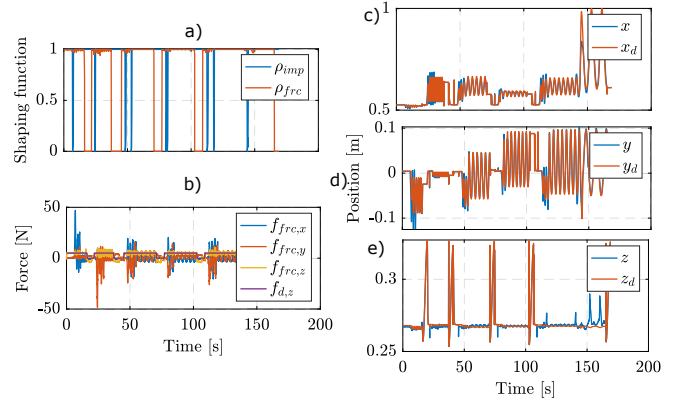


Fig. 7: **Experiment III: Evolution of ρ_{imp} and ρ_{frc} during polishing and adaptation of $f_{frc,z}$ and motion x, y, z .** a,b) The robot autonomously executes $\rho_{imp} = 1$, robustly stops and re-establishes contact via ρ_{frc} . $f_{frc,z}$ smoothly changes from 5 N to 0 N during contact loss, thereafter increases to 5 N with re-establishing contact c-e) set point for contact x_d, y_d, z_d vs. current position of the robot x, y, z during contact, contact loss and re-establishing contact

as well. In future work, we will investigate force learning through stiffness adaptation and the decision-making process for controlling the maximum allowable energy transfer between the robot and its environment during free and contact motions.

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